

LSTM and ARIMA based binary classification problem for Tesla stock rise and fall

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Abstract

The volatility of Tesla's stock prices presents a unique challenge and opportunity for financial forecasting. This study investigates various modeling approaches to predict the directional movement of Tesla's stock prices, leveraging data from November 2014 to November 2024. Traditional stock forecasting models like ARIMA was used to predict the future trend, Then stocks machine learning techniques, including Random Forest and Support Vector Machines (SVM), were applied to classify daily stock movements, achieving modest accuracies of 57% and 53%, respectively. Recognizing the limitations of these methods in capturing temporal dependencies, we implemented a Long Short-Term Memory (LSTM) network, which demonstrated superior performance with an accuracy of 77.8%. Key improvements were achieved by narrowing the training dataset to recent data (2023–2024) and optimizing lag parameters to capture relevant trends. This study highlights the potential of deep learning in financial time series analysis and provides a robust foundation for developing data-driven investment strategies, achieving 78.9% reliability in simulated trading scenarios.

1 Introduction

Tesla, Inc., recognized for its pivotal role in revolutionizing the electric vehicle (EV) and sustainable energy sectors, offers an intriguing subject for financial analysis due to its innovative business model and significant technological advancements. The fluctuating stock prices of Tesla not only reflect its market volatility but also offer a unique opportunity to gauge the impact of technological innovation on financial valuation. Understanding the dynamics of Tesla's stock is crucial for investors who navigate the rapid shifts in technology and market conditions that influence the automotive and energy industries.

In this research, we employ a variety of advanced analytical techniques to dissect the complex behavior of Tesla's stock price. We utilize ARIMA (AutoRegressive Integrated Moving Average) models to capture the linear aspects of time series data, which help in understanding the general trends and seasonality effects.

Alongside, in order to transform the stock price regression problem, into a simple classification problem of predicting whether the stock price will increase or decrease the next day, we integrate machine learning approaches such as Support Vector Machines (SVM) and Random Forests (RF)

[1] to harness their predictive power in capturing nonlinear relationships and complex patterns that are not apparent through traditional methods. They got 53% and 55 %accuracy respectively.

Further, we leverage Long Short-Term Memory (LSTM) networks, a class of deep learning models well-suited for sequential data, to predict Tesla's stock movements based on historical price data. LSTM networks are particularly effective in managing data where long-term temporal dependencies are crucial for forecasting future stock price trajectories and get 77% accuracy in prediction.

The primary aim of this study is to build robust predictive models that can accurately forecast the directional movements of Tesla's stock, focusing particularly on the next day's closing price relative to the previous day. This approach not only aids in refining investment strategies but also provides a deeper insight into the market's reaction to ongoing and anticipated changes within Tesla and the broader EV landscape. Through this multifaceted analytical endeavor, we seek to provide valuable foresights that could assist investors, analysts, and policymakers in making well-informed decisions in a highly volatile market.

2 Related work

Predicting Stock Price Movements. Predicting specific stock prices precisely is inherently challenging due to the volatile nature of financial markets. Studies have shown that forecasting the direction of stock price movements rather than exact future prices can be more practical for investors. Research by Lee and Yoon (2019) demonstrates the effectiveness of predictive analytics focused on price directionality using historical price data and technical indicators, suggesting a strategic approach to stock trading [1].

Influence of Market Sentiment on Price Movements. The role of market sentiment in influencing stock price fluctuations is well-documented in financial research. Johnson et al. (2020) analyzed the impact of news sentiment and social media trends on stocks and found significant correlations between public sentiment and the subsequent price movements, particularly in the technology sector [2]. This indicates the potential of sentiment analysis in improving the accuracy of directional forecasts.

Application of Machine Learning Techniques in Finance. Machine learning techniques are increasingly utilized to predict stock price movements due to their ability to handle large datasets and learn complex patterns. Zhao and Park (2018) implemented various algorithms to predict up or down movements in the stock market with considerable success, highlighting the utility of ensemble methods and deep learning models like Long Short-Term Memory (LSTM) networks in capturing the temporal dynamics of stock prices [3] [5].

Investment Strategies Based on Predictive Models. The practical application of predictive models extends to formulating investment strategies. Turner and Sharma (2021) explore how directional predictions of stock prices facilitate the development of automated trading systems that outperform traditional buy-and-hold strategies, especially in volatile markets. They argue that the

integration of quantitative models into trading strategies can significantly mitigate risk and enhance returns [4].

These studies collectively underscore the challenges and strategies involved in predicting stock price movements, emphasizing the shift from attempting to predict exact prices to forecasting directional changes.

3 Data

In our study, we delve into the intricate dynamics of Tesla's stock price from November 1, 2014, to November 1, 2024, utilizing a dataset sourced from **Yahoo Finance**. This dataset encapsulates a comprehensive view of the fluctuations within the Tesla stock over a decade, thus providing a rich basis for understanding the underlying patterns that may influence its market behavior. The data spans daily trading metrics including open, high, low, close prices, and trading volume, enriched further by integrating broader economic indicators such as the NASDAQ index and Federal Reserve's 10-year Treasury rates.

3.1 Feature Engineering[10]

To refine our analysis and improve the predictive capabilities of our models, extensive feature engineering was conducted:

- **Daily Returns:** Calculated to capture the day-to-day percentage change in the adjusted closing price, providing insights into the volatility and trend momentum of the stock.

$$R_t = \frac{P_{t, \text{adj close}} - P_{t-1, \text{adj close}}}{P_{t-1, \text{adj close}}}$$

Where $P_{t, \text{adj close}}$ is the adjusted close price on day t . This feature calculates the day-to-day percentage change in the stock price, providing insights into the stock's volatility and trend momentum.

- **Lag Features:** Up to 14 days of lagged values of the adjusted close price were included to encapsulate the temporal autocorrelations, which are crucial for capturing the momentum and mean-reversion effects in stock prices.

$$L_{t,k} = P_{t-k, \text{adj close}}$$

Here k ranges from 1 to 14, representing the adjusted close price from k days ago.

These features help capture the temporal autocorrelations and are essential for modeling momentum and mean-reversion effects in stock prices.

- **Moving Averages:** 5-day and 10-day moving averages were computed to smooth out short-term fluctuations and highlight longer-term trends in the stock price.

$$MA_{5,t} = \frac{1}{5} \sum_{k=0}^4 P_{t-k, \text{ adj close}}$$

$$MA_{10,t} = \frac{1}{10} \sum_{k=0}^9 P_{t-k, \text{ adj close}}$$

Moving averages smooth out price data to illustrate clearer market trends by averaging the prices over specific intervals.

- **Relative Strength Index (RSI):** A momentum oscillator used to measure the speed and change of price movements, providing a gauge of price strength and potential reversals.

$$RSI_t = 100 - \frac{100}{1 + RS_t}$$

Where:

$$RS_t = \frac{\text{average gains over N days}}{\text{average losses over N days}}$$

- **Moving Average Convergence Divergence (MACD):** Utilized to reveal changes in the strength, direction, momentum, and duration of a trend in the stock's price.

$$MACD = EMA_{12} - EMA_{26}$$

- **Daily Range and Close/Open Difference:** These features capture daily price volatility and the intraday price movement relative to the opening price, respectively.

$$D_{t,co} = P_{t,close} - P_{t,open}$$

These features measure daily price volatility and the difference between closing and opening prices, offering insights into intraday price movements.

- **Normalized Prices:** These features standardize the closing and opening prices by the daily range, providing a normalized perspective of price movements.

$$\text{Norm}_{t, \text{close}} = \frac{P_{t, \text{close}}}{D_{t, \text{range}}}$$

$$\text{Norm}_{t, \text{open}} = \frac{P_{t, \text{open}}}{D_{t, \text{range}}}$$

Normalized prices adjust the closing and opening prices relative to the daily price range, providing a scaled perspective of price movements.

4 Model

4.1 ARIMA Time Series Model

This time series analysis initially examines a decade of Tesla's stock prices, revealing significant fluctuations over the years, indicative of high market volatility and non-stationarity in the data.

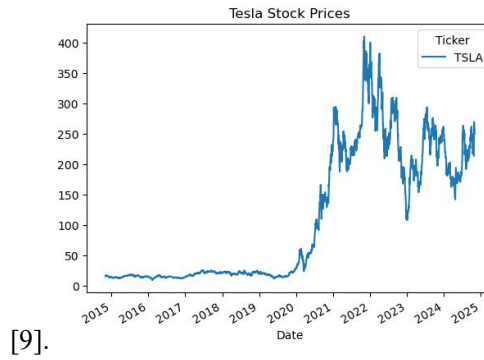


Figure1: 10-year time series for Tesla Closing Price

To address this, a focused dataset from 2022 to 2024 was chosen for its series stability. Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots were utilized to determine the appropriate order for an ARIMA model.

The autocorrelation function (ACF) and partial autocorrelation function (PACF) plots displayed in Figure 2 are instrumental in determining the order of our ARIMA model. The ACF plot exhibits a sharp cutoff after the first lag, suggesting that the series reaches non-significance quickly, which typically indicates an MA component of order 1. Similarly, the PACF plot shows a tailing off after the first lag, suggesting an AR component of order 1.

Given these observations, the ARIMA model selected for modeling Tesla's stock prices is an ARIMA(1,1,1). This model incorporates:

$$\nabla X_t = \alpha + \phi_1 \nabla X_{t-1} + \theta_1 \epsilon_{t-1} + \epsilon_t$$

where:

∇X_t is the first difference of the series at time t ,

α is a constant term,

ϕ_1 is the coefficient for the first autoregressive term,

θ_1 is the coefficient for the first moving average term,

ϵ_t is the error term at time t

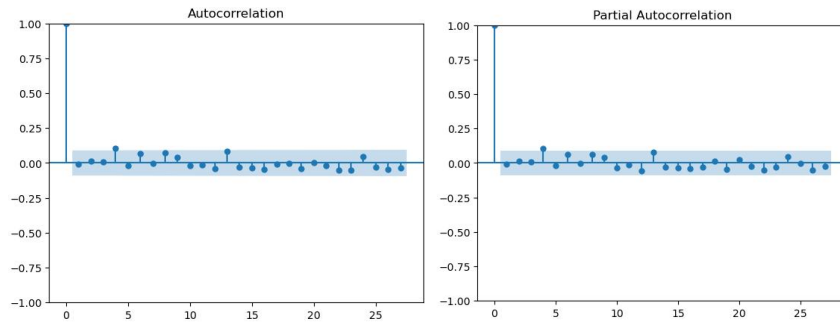


Figure 2: ACF and PCF for ARIMA model

However, the selected model was unable to achieve a stable time series analysis, largely due to the high volatility of the stock prices, which led to unreliable forecasts. Consequently, the focus of our research shifted from a regression model predicting precise future values to a classification model that predicts whether the stock price will rise or fall compared to the previous day. This approach more effectively addresses the challenges posed by the unpredictable nature of financial time series.

4.2 Random Forest Model

To transform the problem in to a classification problem, a RandomForestClassifier was used to predict the next day's stock price movement (up or down)[1]. The model was trained using various time ranges of historical data, starting with 10 years (2014–2024) and progressively narrowing to 1 year (2023–2024).

Through experimentation, i discovered that using 3 years of data (2021–2024) for training yielded the highest prediction accuracy of 54%. This improvement is likely due to Tesla's significant stock price surge in 2020–2021 from **Figure1**, which rendered the earlier 7 years of data less relevant for the current market dynamics.

After building the Random Forest model, we analyzed feature importances to identify the variables that contributed most to the model's performance. This allowed us to assess the influence of each feature on the model's predictive power.

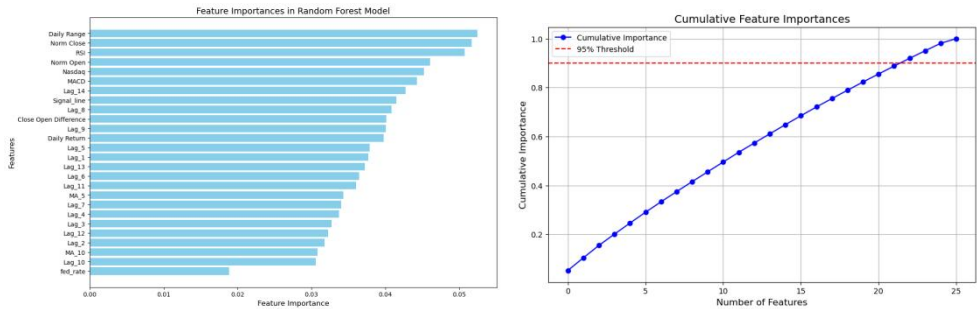


Figure 3: Feature importance in Random Forest Model

From the feature importance chart **Figure 3**, the following top 6 features significantly impact the Random Forest model's prediction of Tesla stock price movements: **Daily Range**, Norm Close, RSI, Norm Open, Nasdaq Index, MACD.

These features indicate that Tesla's stock price movement is heavily influenced by daily price volatility (Daily Range, Norm Close, Norm Open), market trend indicators (Nasdaq and MACD), and momentum oscillators (RSI). Together, these features reflect market sentiment, trend strength, and price dynamics, forming the foundation for robust predictive modeling.

Based on the ranked feature importances, we set 95% cumulative importance line to remove less relevant variables and retrained the model with a reduced feature set. The simplified model showed an improved prediction accuracy of 57%. This indicates that reducing noise and focusing on the most critical features can enhance the model's performance.

Then I implement a 5 fold cross-validation to my model and use predict accuracy as score, which shows that the training score (red line) is consistently at 1.0, indicating that the model perfectly fits the training data. However my cross-validation score (green line) remains significantly lower than the training score, hovering around 0.5–0.6 which is a not good results for a binary classification models, only a little better than a coin flip.

This results suggest that the model is overfitting the training data, as it memorizes the training examples rather than generalizing well. Model did not learn real rules about how to judge rise or fall in price of stocks.

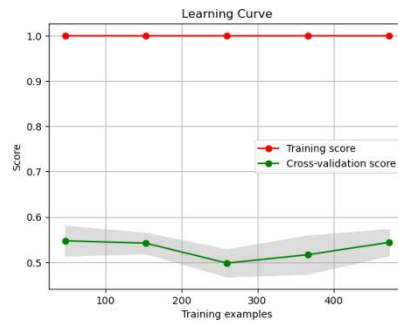


Figure 4: Learning curve for Random Forest

4.3 SVM Model

After observing the suboptimal performance of the Random Forest model, we turned to a Support Vector Machine (SVM) model in an effort to improve prediction accuracy. Given the inherent complexity and volatility of stock price movements, we hypothesized that the non-linear patterns within the data might be better captured by SVM. To this end, we employed the Radial Basis Function (RBF) kernel, which is particularly adept at identifying non-linear relationships by mapping data into a higher-dimensional space.

To optimize the model, we conducted a thorough grid search combined with 10-fold cross-validation to identify the best hyperparameters, including the regularization parameter (C) and the kernel coefficient (γ). The SVM model with the optimal parameters only obtained a 53% accuracy. Despite these efforts, while the SVM model with the RBF kernel demonstrated no improvement over the Random Forest model.

As the learning curve shows, the training score initially starts high, reflecting the model's ability to fit well to smaller training sets. However, as the training set size increases, the training score slightly decreases, indicating that the model is adjusting to capture broader patterns in the data. On the other hand, the cross-validation score remains consistently lower and relatively flat, suggesting limited generalization capability. The overall prediction accuracy remained modest, reflecting the inherent challenges of modeling stock price trends.

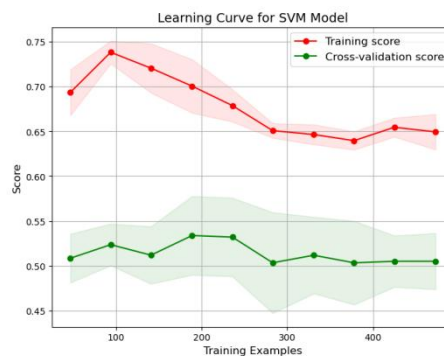


Figure 5: Learning curve for SVM

4.4 LSTM Model

Stock price prediction is challenging due to the volatile, noisy, and non-linear nature of financial time-series data. Traditional models like Random Forest and SVM struggle to capture temporal dependencies and long-term patterns inherent in such data, motivating the use of Long Short-Term Memory (LSTM) networks.

LSTMs, a type RNN, are designed to handle sequential data by retaining relevant information over time while filtering out noise. This makes them well-suited for Tesla's stock prediction, where prices are influenced by both short-term events (e.g., earnings reports) and long-term factors (e.g., technological advancements)[5].

Previous studies have demonstrated LSTM's superiority in financial applications. For example, Bao et al. (2017) [6] applied LSTM to financial time series and achieved notable improvements in prediction accuracy compared to traditional models. Similarly, Fischer and Krauss (2018)[7] showed that LSTMs outperformed traditional machine learning methods in financial markets

In our model, I experiment and decide to use parameters below:

Parameter	Value	Reason
LSTM Layer number	50	balance between complexity and computational cost
Dropout Rate	0.2	to prevent overfitting
Activation Function	Sigmoid	for binary classification
Optimizer	Adam	known for its adaptive learning rate
Loss Function	Binary cross-entropy	suitable for classification tasks
Batch Size	32	balance between computational efficiency and model convergence
Epochs	50	with early stopping applied to avoid overfitting
Early Stopping Patience:	10	to monitor validation loss
Time Steps (Lag)	1:200	Explored values ranging from 1 to 200, with optimal results obtained at a specific lag

Table 1: Parameters for LSTM Model

In our research, we explored multiple time spans to optimize the performance of our predictive model. Starting with the full dataset from 2014 to 2024, we gradually reduced the time range and found that focusing on the most recent year, 2023-2024, significantly improved the model's performance. Specifically, the accuracy increased by 25% compared to using the entire 10-year range. Furthermore, we tested various lag values to capture temporal dependencies effectively. Ultimately, we identified that using a lag of 197 yielded the best results, achieving an impressive prediction accuracy of **77.8%**. This demonstrates the importance of narrowing the dataset to capture the most relevant trends and fine-tuning hyperparameters like lag to enhance the model's predictive power.

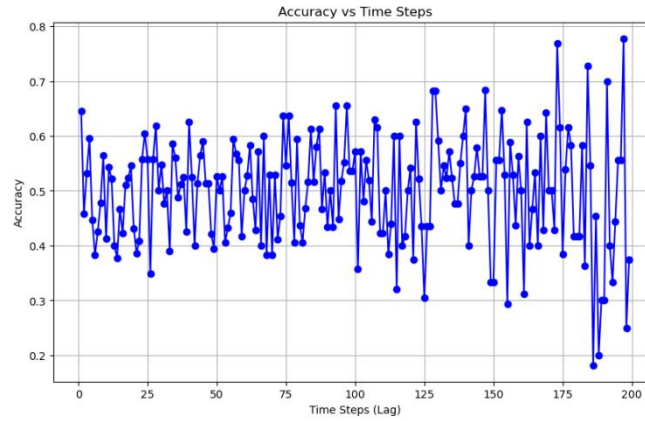


Figure 6: The training accuracy for different time lag

5 Conclusion and Strategy

Conclusion

This study explored various predictive modeling approaches to forecast Tesla's stock price movements, including Random Forest, SVM, and LSTM. While traditional models like Random Forest and SVM achieved modest accuracy of 57% and 53%, respectively, the LSTM model demonstrated its ability to handle the volatile and sequential nature of financial data, achieving a remarkable prediction accuracy of 77.8%. The improvement came from narrowing the training dataset to the most recent year (2023–2024) and fine-tuning the lag parameter to 197, effectively capturing temporal dependencies.

Model	Accuracy	MSE	Data Time Span
Random Forest	0.57	0.4544	2021.11-2024.11
SVM	0.53	0.4600	2021.11-2024.11
LSTM	0.778	0.134	2023.11-2024.11

Table 2 : Comparison of three models

The results show that the LSTM model has better performance than other models in predicting short-term stock price movements. Our analysis shows that stock price prediction benefits significantly from models capable of capturing short-term patterns and handling noisy, volatile data. The LSTM model's success underscores its suitability for complex financial time series compared to traditional methods. However, stock prices are highly susceptible to economic, market and political factor.[8]

Strategy

If the model is indeed stable and Tesla has a smooth stock price sequence in the recent past, with an LSTM model with such a high prediction rate, I built a simple investment strategy based on it with geometric expectations.

Core Idea:

Divide Capital Over Four Days:

- The investment capital is evenly divided across four consecutive trading days. For example, if the total capital is \$100, then \$25 is allocated each day.
- This ensures that the risk is distributed over multiple trading days, reducing the impact of potential losses.

Action Based on Model Prediction:

- If the model predicts the stock price will rise the next day, a **long position** is taken (buying the stock or long options).
- If the model predicts the stock price will fall, a **short position** is taken (selling the stock short through mechanisms like margin trading or buying short options).

The availability of this strategy:

With a model accuracy of 77.8%, the probability of all four predictions being correct is:

$$P(\text{ All Correct }) = (0.778)^4 = 0.365$$

- The probability of at most one incorrect prediction over the four days is:

$$P(\text{ At Most 1 Error }) = P(\text{ All Correct }) + P(1 \text{ Err or }) = (0.778)^4 + \binom{4}{1} \cdot (0.778)^3 \cdot (0.222)^1 = 0.789$$

Thus, there is a 78.9% probability of achieving a positive return across the four-day investment period. It is certain that the exact amount of investment and the multiples of long and short to ensure that the risk 'evenly divided' still need to be measured in the actual situation, but this is a easy and viable strategy for my model.

Code Availability

<https://colab.research.google.com/drive/1u0ayQvwnhay-8w6AxL8nvZfIbdBfgyNI>

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